

Supporting Information for:

**Cytochrome c_{552} Mutants: Structure and Dynamics at the Active Site Probed by
Multidimensional NMR and Vibration Echo Spectroscopy**

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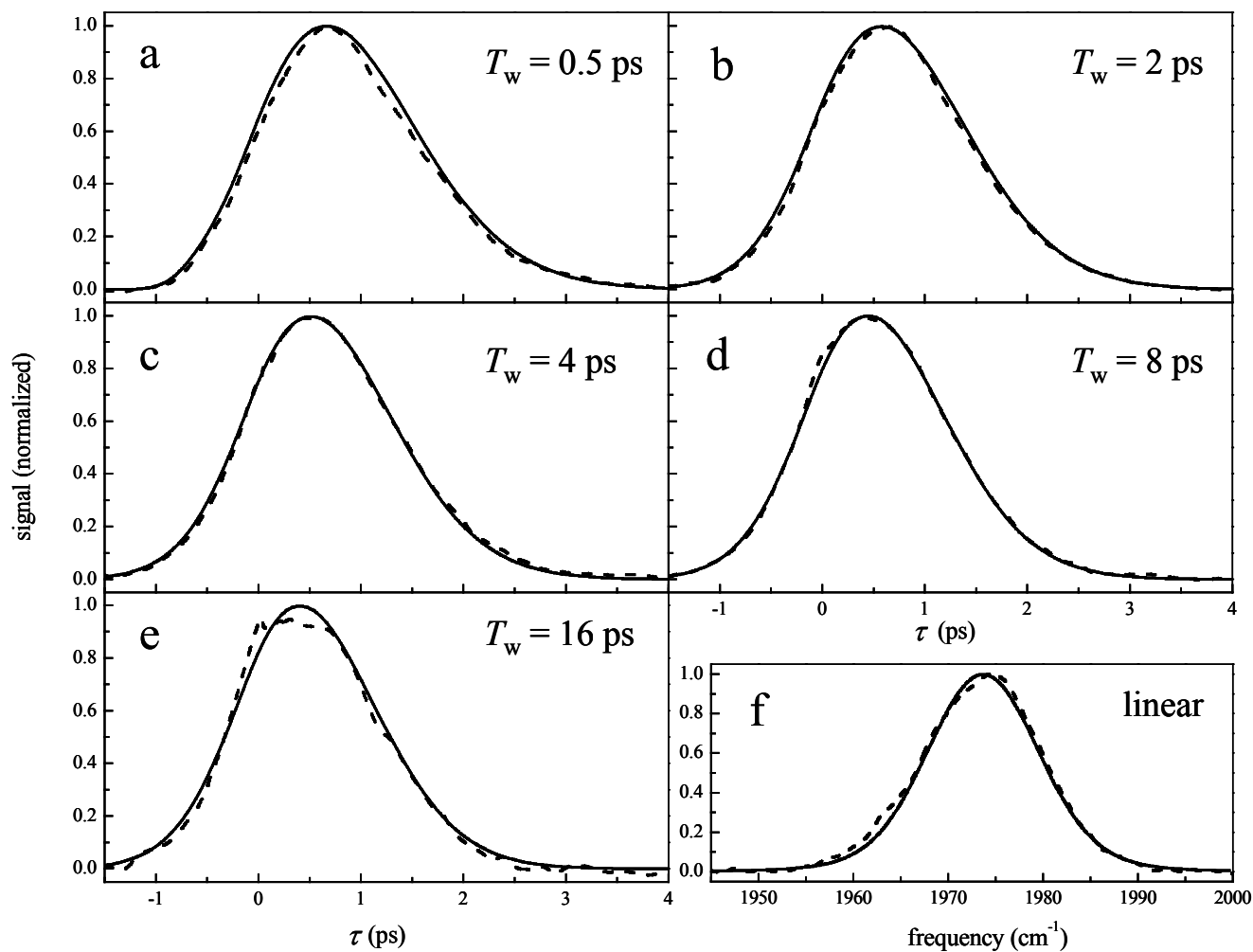


Figure S1. a) Experimental vibrational echo decay data at T_w = a) 0.5 ps, b) 2 ps, c) 4 ps, d) 8 ps, e) 16 ps, and f) linear spectrum for *Ht*-M61A (dashed lines) overlaid with the best-fit vibrational echo decay and linear spectrum calculated from nonlinear response theory (solid lines) at 1975 cm⁻¹.

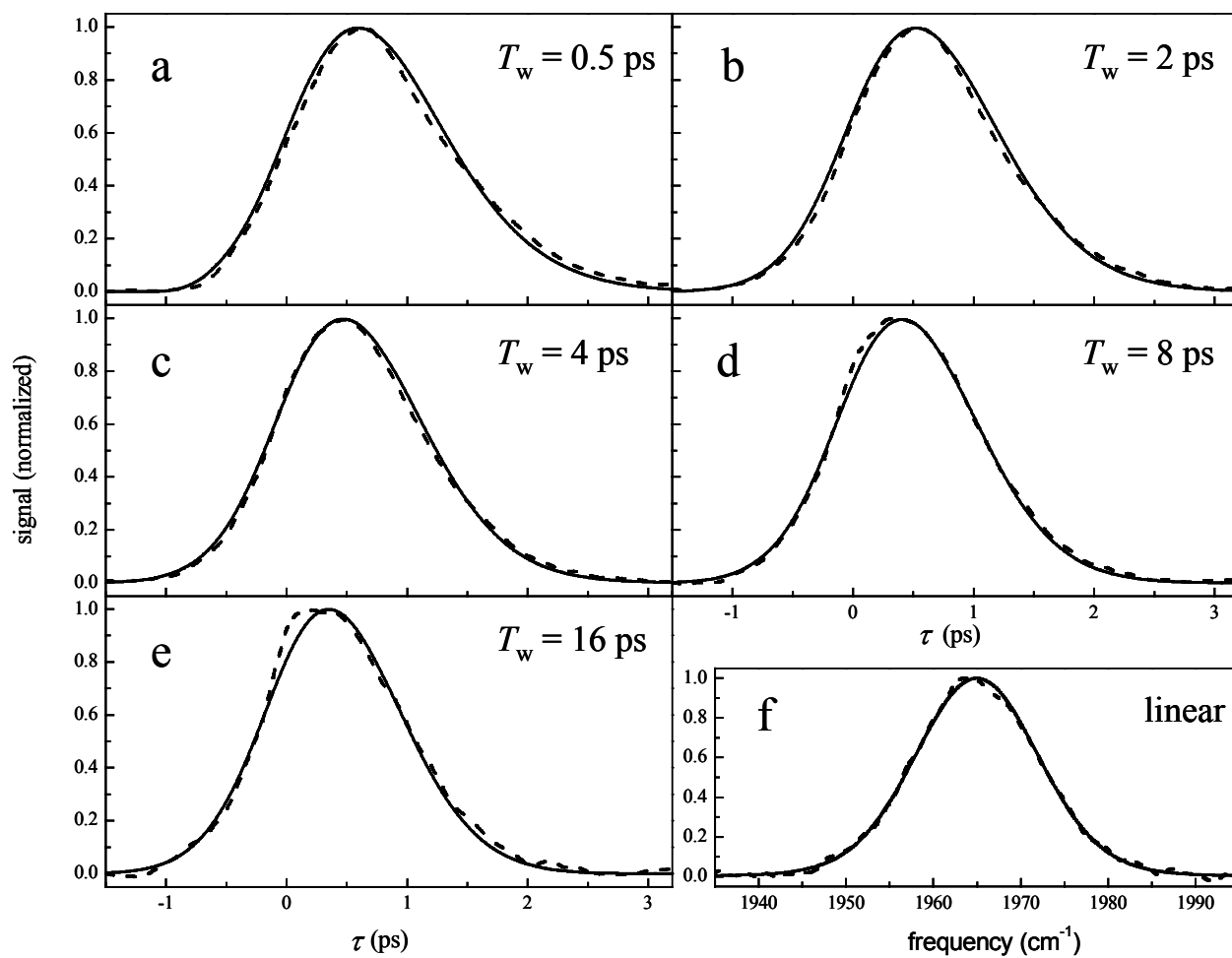


Figure S2. a) Experimental vibrational echo decay data at $T_w =$ a) 0.5 ps, b) 2 ps, c) 4 ps, d) 8 ps, e) 16 ps, and f) linear spectrum for *Ht*-M61A/Q64N (dashed lines) overlaid with the best-fit vibrational echo decay and linear spectrum calculated from nonlinear response theory (solid lines) at 1965 cm^{-1} .